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JIANPING GOU, Southwest University, Chongqing, China

LIYUAN SUN, Jiangsu University, Zhenjiang, Jiangsu, China

BAOSHENG YU, The University of Sydney, Sydney, NSW, Australia

SHAOHUA WAN, University of Electronic Science and Technology of China, Chengdu, Sichuan, China

DACHENG TAO, The University of Sydney, Sydney, NSW, Australia

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Hierarchical Multi-Attention Transfer for Knowledge Distillation

JIANPING GOU, College of Computer and Information Science, College of Software, Southwest University, China
School of Computer Science and Communication Engineering, Jiangsu University, China
LIYUAN SUN, School of Computer Science and Communication Engineering, Jiangsu University, China
BAOSHENG YU, School of Computer Science, The University of Sydney, Australia
SHAOHUA WAN, Shenzhen Institute for Advanced Study, University of Electronic Science and Technology of China, China
DACHENG TAO, School of Computer Science, The University of Sydney, Australia

Knowledge distillation (KD) is a powerful and widely applicable technique for the compression of deep learning models. The main idea of knowledge distillation is to transfer knowledge from a large teacher model to a small student model, where the attention mechanism has been intensively explored in regard to its great flexibility for managing different teacher-student architectures. However, existing attention-based methods usually transfer similar attention knowledge from the intermediate layers of deep neural networks, leaving the hierarchical structure of deep representation learning poorly investigated for knowledge distillation. In this paper, we propose a **hierarchical multi-attention transfer framework (HMAT)**, where different types of attention are utilized to transfer the knowledge at different levels of deep representation learning for knowledge distillation. Specifically, position-based and channel-based attention knowledge characterize the knowledge from low-level and high-level feature representations, respectively, and activation-based attention knowledge characterize the knowledge from both mid-level and high-level feature representations. Extensive experiments on three popular visual recognition tasks, image classification, image retrieval, and object detection, demonstrate that the proposed hierarchical multi-attention transfer or HMAT significantly outperforms recent state-of-the-art KD methods.

CCS Concepts: • **Computing Methodologies** → **Artificial Intelligence**; *Deep Learning*;

Additional Key Words and Phrases: Model compression, knowledge distillation, hierarchical attention transfer

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Authors' addresses: J. Gou (corresponding author), College of Computer and Information Science, College of Software, Southwest University, Chongqing 400715, PR China and also School of Computer Science and Communication Engineering, Jiangsu University, Zhenjiang, Jiangsu 212013, China; email: cherish.gjp@gmail.com; L. Sun, School of Computer Science and Communication Engineering, Jiangsu University, Zhenjiang, Jiangsu 212013, PR China; email: 2211908031@stmail.ujs.edu.cn; B. Yu and D. Tao, School of Computer Science, the Faculty of Engineering, The University of Sydney, Sydney, NSW 2008, Australia; emails: {baosheng.yu, dacheng.tao}@sydney.edu.au; S. Wan, Shenzhen Institute for Advanced Study, University of Electronic Science and Technology of China, Shenzhen, 518110, PR China; email: shaohua.wan@ieee.org.

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1 INTRODUCTION

In recent years, deep neural networks have achieved very impressive performance in vision, speech and language applications, while typical deep neural networks usually contain hundreds of layers and millions of parameters. Therefore, a subsequent consequence is that the model complexity has dramatically increased with the number of layers, making it a great challenge for the model deployment on mobile and embedded devices. Since then, a variety of model compression techniques have been proposed to learn efficient and lightweight models, including network pruning [14, 27], model quantization [20, 35], lightweight architectures [18, 29, 37], and knowledge distillation [1, 4, 17, 23, 33, 53, 55]. Specifically, knowledge distillation is very easy to implement in conjunction with other model compression methods, thus it has attracted widespread research interests from the community.

Feature representations from the intermediate layers of deep neural networks have been widely used to transfer knowledge from teacher to student models [1, 2, 33, 46, 54]. Among feature-based knowledge transfer methods, attention-based knowledge has received increasing attention due to not only good performance but also its great flexibility for teacher-student learning architectures [5, 21, 23, 34, 43, 52]. Specifically, attention-based knowledge transfer does not require a perfect match between teacher and student feature maps, including the spatial dimensions and the number of channels. However, existing attention-based knowledge distillation methods usually utilize the same attention knowledge from different intermediate layers, leaving the hierarchical structure of deep neural networks for representation learning poorly investigated. Deep neural networks consist of a hierarchy of information such as edges/corners in early layers and object parts in middle/deep layers, as shown in Figure 1. Obviously, Figure 1 visually shows that the neural network focuses on different parts of the object in different middle layers. To be specific, the shallow layers of the neural network mainly extract the partial information and local parts of the object, and the deeper layers of neural networks can focus on the whole information and global parts of the object. Therefore, it is of great importance to explore different types of attention knowledge in regard to low-level, middle-level, and high-level feature representations. Inspired by this, we devise a new knowledge distillation framework, hierarchical multi-attention transfer or HMAT, which effectively transfers different types of attention knowledge from different intermediate layers of deep neural networks during the teacher-student learning.

In the proposed HMAT framework for knowledge distillation, we consider three different kinds of attention-based knowledge, namely activation-based attention knowledge [23], channel-based attention knowledge [34], and position-based attention knowledge [18]. Specifically, the main motivations are as follows: (1) considering that deep neural networks usually learn to detect low-level features like edges in the early layers while learning, the position-based attention knowledge is thus used to transfer the knowledge from the early layers of deep neural networks; (2) middle/deep layers progressively combine features of earlier layers to form a more holistic representation. As shown in the last column of Figure 1, we visualize the attention map of the last block in deep neural network, where the whole object is highlighted, suggesting that deep layers usually focus on learning global abstract features. Therefore, we adopt the widely used activation-based attention knowledge to mimic deep layers of the teacher model; and (3) the activation-based attention

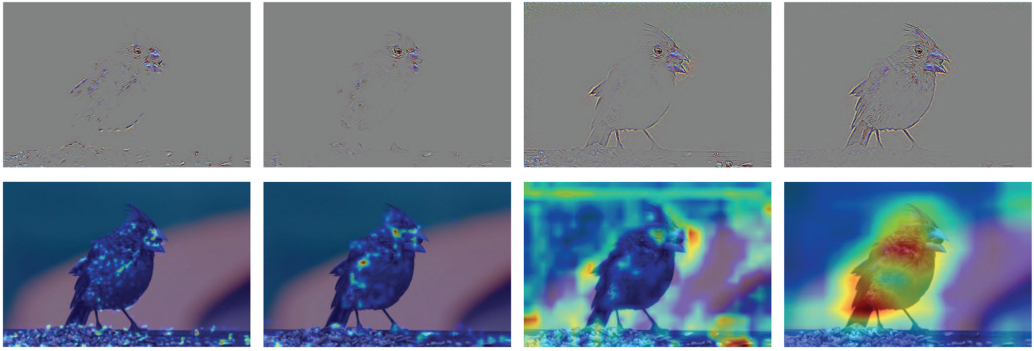


Fig. 1. An intuitive example of the hierarchy of information in convolutional neural networks from low to high levels. Specifically, the activations from early layers usually focus on semantic details such as edges/corners, while deep layers focus on the whole objects.

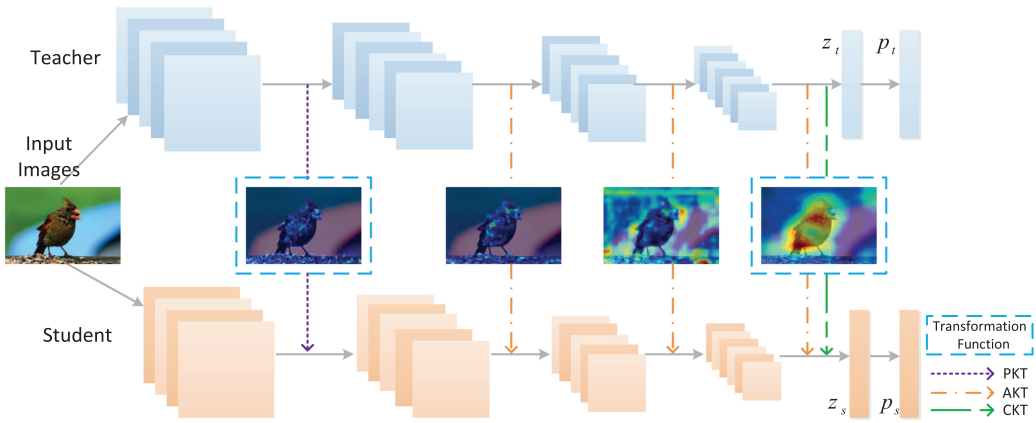


Fig. 2. The main hierarchical multi-attention transfer framework (HMAT) for knowledge distillation. Note that the transformation function is a combination of convolutional and global average pooling operations for **channel-based attention knowledge transfer (CKT)**, a combination of convolutional, concatenation and activation operations for **position-based attention knowledge transfer (PKT)**. **AKT** indicates the **activation-based attention knowledge transfer**. z_t and z_s are logits of teacher and student networks, respectively, and p_t and p_s are their predictions (i.e., soften logits).

knowledge are usually derived from the feature maps along the channel dimension (e.g., average/max pooling), while per-channel knowledge is missing. As a result, we further introduce the channel-based attention knowledge from the deepest layer to improve the performance of the student model. The main HMAT framework for knowledge distillation is illustrated in Figure 2. To our best knowledge, this is the first multi-attention knowledge transfer framework that explores different kinds of attention knowledge in different intermediate layers of deep neural networks to effectively guide the learning of student models.

The proposed method aims to provide more informative guidance from different teacher-model layers to the student model in a hierarchical manner. To evaluate the proposed method for knowledge distillation, we conduct a number of experiments on three important visual recognition tasks using seven popular image datasets. Experimental results demonstrate the superior performance of

the proposed HMAT framework over recent state-of-the-art KD methods. The main contributions of this paper can be summarized as follows:

- A new hierarchical multi-attention transfer framework for knowledge distillation: It transfers different attention knowledge in different intermediate layers of the neural network.
- Different types of attention knowledge are explored to provide insights on which kind of attention knowledge is appropriate to transfer in the different layers of the teacher model.
- A new position-based attention knowledge is devised for the proposed hierarchical multi-attention transfer.
- Extensive experiments and ablation studies on a variety of teacher-student learning networks: we compare HMAT with recent state-of-the-art methods on three different visual recognition tasks, including image classification, image retrieval, and object detection, to demonstrate the effectiveness of HMAT.

The rest of this paper is organised as follows. In Section 2, we first review existing attention-based knowledge distillation methods. Section 3 then introduces the proposed HMAT framework for knowledge distillation. Experimental results including ablation studies are shown in Section 4 to demonstrate the effectiveness of the proposed method. Finally, we conclude this paper in Section 5.

2 RELATED WORK

In this section, we first briefly review knowledge distillation, especially feature-based knowledge distillation. We then introduce typical attention-based knowledge transfer as well as the channel-wise knowledge in deep neural networks.

2.1 Knowledge Distillation

Due to the excellent performance of deep neural networks in vision, speech and language tasks, various knowledge distillation schemes [1, 12, 17, 28, 33, 41, 44, 46, 47, 54] have been proposed to obtain a high-performance model with fast-running speed to meet the requirements of industrial production. To obtain better performance of the compact model, different distillation frameworks and different types of knowledge have been intensively explored [12]. Since FitNet [1] was proposed, more and more work began to study the knowledge in the middle layers of neural network. [45] added auxiliary classifiers in some hidden layers of the teacher network and the student network, then trained the teacher network and the student network collaboratively, and removed these classifiers after training to reduce the running time. [13] proposed a two-stage differentiable feature aggregation search method. The first stage searches for appropriate feature aggregation weights, and the second stage uses the aggregations to perform feature distillation between the teacher network and the student network. [16] proposed a novel feature distillation framework where the distillation loss makes a synergy among teacher and student transformation, distance function, and distillation feature position. [11] proposed a bridge distillation framework, significantly reducing the memory requirements and the computational cost. [50] proposed an evolutionary method to promote the student learning by narrowing the ability gap between teacher and student networks. [54] proposed a promising feature map distillation method, which effectively learns compact student network for low-resolution object recognition with high performance.

2.2 Attention Knowledge

In recent years, attention mechanism has become a research hotspot in vision, speech and language tasks, including machine translation [3], speech recognition [7], semantic segmentation [40], and image classification [23, 34, 52]. The attention mechanism provides a model with the ability to make more accurate prediction without significantly incurring additional computation and storage

overhead. Researchers have proposed many methods to visualize the attention maps of neural networks [30, 38, 49]. Among them, Grad-CAM [38] has attracted extensive attention due to its good performance and convenient application.

Subsequently, [23] proposed a standard **attention transfer (AT)** for knowledge distillation. Specifically, in the training process, the authors used activation-based or gradient-based attention maps for attention transfer. The authors also verified through experiments that using activation-based attention maps for attention transfer has better performance than using gradient-based attention maps. Since then, activation-based attention knowledge has been explored and applied in the field of knowledge distillation. In [52], the framework of multiple teachers teaching one student is adopted, and one of the teacher networks is defined as an expert teacher to transfer activation-based attention knowledge to the student. [34] adopted two different attention mechanisms to model two kinds of knowledge including activation-based attention knowledge and channel-based attention knowledge for knowledge transfer. [26] fused two model compression mechanisms of knowledge distillation and pruning, and transferred the activation-based attention knowledge to the student network during distillation process. Similar to attention knowledge transfer, the knowledge from the feature maps is transformed into transportable knowledge via factor transfer, so as to make the student model easily understand the teacher model [22]. Note that the attention maps are usually formed by the transformation functions such as the convolutional, concatenation and activation operations, while the feature maps in factor transfer are learned by adding a small network with additional computational cost. Recently, in [51], the authors have proposed attention-guided distillation method. The attention mechanism enables the student model to focus more on the crucial pixels and channels instead of the whole feature maps.

In attention transfer for knowledge distillation, attention maps are usually obtained by transforming the feature maps of a middle layer, leaving the knowledge from multi-level layers poorly investigated. To address the above-mentioned issues, we further explore the feature knowledge from different levels via a multi-attention mechanism, and thus devise a hierarchical multi-attention knowledge transfer framework.

2.3 Channel Knowledge

Considering the rich knowledge in channel dimensions of deep neural networks, a variety of channel-based knowledge thus have been introduced for knowledge distillation. For example, [31] explored the optimal structure of the network by training channels, and [39] proposed a novel channel-wide distillation scheme for semantic segmentation and achieved very good performance. Recently, several methods using channel-based attention knowledge have been developed [6, 34]. Specifically, [34] proposed a hybrid attention transfer method for knowledge distillation, where the channel-wise attention is used as a supplement of the activation-based attention. [6] devised a novel SCA-CNN to incorporate spatial-wise attention and channel-wise attention for knowledge transfer.

Existing methods have demonstrated that the knowledge from different channels is very favorable for knowledge distillation. However, none of them considers whether different attention maps including channel-based attention map are suitable or not for knowledge transfer in different layers. Therefore, we devise a novel framework to enable the student network to learn proper attention from different middle layers of the teacher network, and thus improve the performance of teacher-student learning.

In summary, different from these related methods aforementioned above, our proposed HMAT method explores multi-attention knowledge transfer with different attention-based knowledge in a hierarchical way. Based on the argument that different layers of the neural network focus on different regions of the object, we properly adapt different attention mechanisms to characterize the

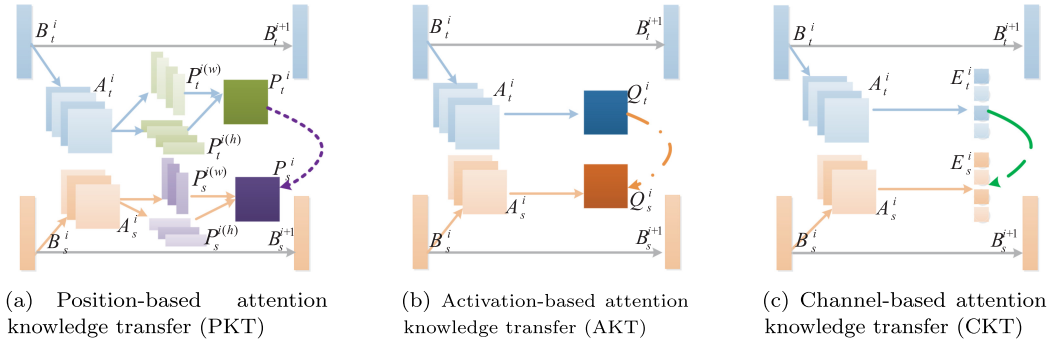


Fig. 3. The detailed transfer process of activation-based attention knowledge, channel-based attention knowledge, and position-based attention knowledge in the distillation process. B_s^i refers to the i -th block of the student network, and B_t^i refers to the i -th block of the teacher network. A_s^i and A_t^i refer to the feature maps of the output of the i -th block of the student and teacher networks. P , Q , and E are different attention maps calculated with the output feature maps, and their specific calculation formulas will be described in the main text.

corresponding attention knowledge from different layers. Through the hierarchical multi-attention knowledge to be distilled, different layers of the student network can learn the informative knowledge from the teacher, so as to better improve its own performance.

3 THE PROPOSED HMAT

In this section, we first introduce the main hierarchical multi-attention transfer framework for knowledge distillation. We then describe proper knowledge transfer in different layers via different types of attention knowledge, i.e., position-based, activation-based, and channel-based attention knowledge. Lastly, we introduce the overall loss function to train very efficient student models using the proposed method.

3.1 Overview

The multi-level layers of the deep neural network have rich semantic information and an intuitive example is shown in Figure 1. With the purpose of distilling proper knowledge from the teacher model into the student model, we focus on extracting these informative multi-level knowledge simultaneously during the teacher-student learning. However, how to effectively model the multi-level knowledge remains a challenge for knowledge distillation. Specifically, in feature-based knowledge distillation, attention transfer has turned out to be an effective strategy [23, 34, 52], while existing attention transfer methods usually adopt only one attention mechanism to distill the feature maps in deep neural networks. To determine a proper attention transfer among multi-level layers, we explore the position-based attention [18], the activation-based attention [23, 34, 52], and the channel-based attention [34] mechanisms for the knowledge transfer in different layers. An intuitive example of different types of attention knowledge transfer in HMAT is shown in Figure 3. Figure 3(a) illustrates the process of transferring position-based attention knowledge described in Section 3.2, which superimposes feature maps along dimensions W and H and allows the student network to learn the position information of the objects concerned by the teacher network. Figure 3(b) illustrates the process of transferring activation-based attention knowledge described in Section 3.3, which superimposes feature maps along dimensions C . Figure 3(c) illustrates the process of transferring channel-based attention knowledge described in Section 3.4, which enables the student network to learn the channel information of the teacher network. Furthermore, to better understand the influence of different attention mechanisms for transferring knowledge in

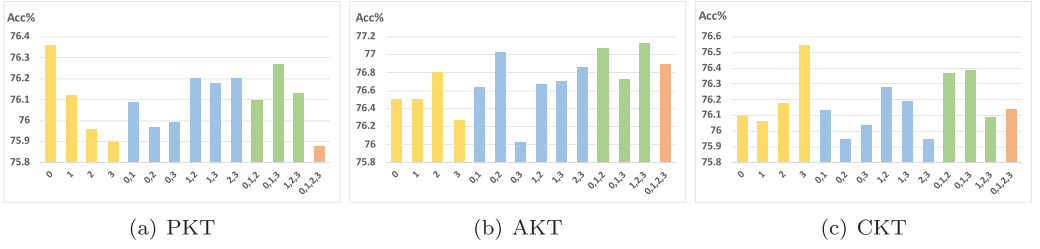


Fig. 4. The KD performance of one type of attention knowledge transfer in certain layers. Note that abscissas 0, 1, 2, and 3 respectively indicate the first, second, third and fourth blocks of the backbone networks including ResNet34 (teacher) and ResNet18 (student) [15]. And yellow, blue, green, and orange means the classification performance that is obtained by transferring knowledge from one block, any two blocks, any three blocks and four blocks of the ResNet model, respectively.

different layers of deep neural networks, we also perform a small grid search using different attention knowledge in different layers, as shown in Figures 4(b), 4(c), and 4(d), respectively. Obviously, the certain layers/blocks should transfer suitable attention knowledge for good performance.

During the process of model training, the proposed HMAT effectively enables various attention knowledge transfer to the certain layers/blocks in a unified framework. When training the student model, different layers can receive a variety of attention knowledge from the teacher network. Therefore, HMAT simultaneously distills different attention knowledge from different intermediate layers of the teacher network to the student network. Specifically, HMAT transfers three different kinds of attention knowledge in the distillation process, namely **position-based attention knowledge transfer (PKT)**, **activation-based attention knowledge transfer (AKT)**, and **channel-based attention knowledge transfer (CKT)**. As shown in Figure 2, three different kinds of attention knowledge transfer can work in a complementary manner. Under the multiple attention knowledge transfer framework, different types of attention knowledge can be effectively integrated to provide more informative supervision for learning very efficient student models.

3.2 Position-based Attention Knowledge Transfer

Given a teacher model T and a student model S , let A_s^i and A_t^i denote the i -th block features of the student and the teacher networks, respectively. In addition, $A \in R^{C \times H \times W}$ consists of C feature planes with $H \times W$ spatial dimensions. Deep neural networks usually pay attention to specific details such as edges and locations at the low abstraction level, while the high-level abstraction representations are not very sensitive to the specific location of the object. Inspired by this, we thus only transfer the position-based attention knowledge at the early layers/stages of deep neural networks. To develop very efficient deep neural networks for mobile devices, [18] embeds the channel information and the positional information into different network block structures. Different from [18], the proposed method devises position-based attention knowledge for knowledge distillation. The feature extraction methods along two spatial directions, W and H , for the student network are as follows:

$$P_s^{i(w)} = \frac{1}{W} \sum_W A_s^i, \quad (1)$$

$$P_s^{i(h)} = \frac{1}{H} \sum_H A_s^i, \quad (2)$$

where $\sum_W$ and $\sum_H$ indicate the accumulation along spatial dimensions. The above two formulas superimpose the feature maps along two spatial directions to generate a new direction-aware

feature maps, which we name as the position-based attention maps. We then concatenate and fuse $P_s^{i(w)}$ and $P_s^{i(h)}$ by using a shared 1×1 convolutional transformation function F as

$$P_s^i = \theta\left(F\left([P_s^{i(h)}, P_s^{i(w)}]\right)\right), \quad (3)$$

where $[\cdot, \cdot]$ indicates the concatenation operation along the spatial dimension and $\theta(\cdot)$ is a non-linear activation function. P_s^i obtained by the above formula is the feature map that encodes spatial information in both the horizontal and vertical direction. Let P_t^i denote the feature map of the teacher network, which is evaluated in a similar way to the above-mentioned student network. For example, when using ResNet18 with the batch-size 128, the size of P_s^i , $P_s^{i(h)}$ and $P_s^{i(w)}$ are $[128, 64, 32, 32]$, $[128, 64, 32, 1]$ and $[128, 64, 1, 32]$, respectively. Therefore, the novel loss function for position-based attention knowledge transfer can be defined as follows:

$$L_{PKT} = \sum_{i \in I} \left\| \frac{P_s^i}{\|P_s^i\|_2} - \frac{P_t^i}{\|P_t^i\|_2} \right\|_2. \quad (4)$$

As argued in [23], l_2 -normalization to the attention maps P in Equation (4) is favorable for the student learning.

3.3 Activation-based Attention Knowledge Transfer

In middle layers of deep neural networks, it usually focuses on local and detailed features, such as eyes and noses, while deep layers pay more attention to global abstract features, such as the entire face. As introduced in [23], when attention transfer is carried out, the student network not only aims to make correct predictions, but also to have similar attention maps with the teacher network. In HMAT, we thus transfer activation-based attention knowledge in the middle and high levels so that the student network can mimic the attention maps from the teacher network as well as possible. The loss function for activation-based attention knowledge transfer can be defined as follows:

$$L_{AKT} = \sum_{i \in I} \left\| \frac{Q_s^i}{\|Q_s^i\|_2} - \frac{Q_t^i}{\|Q_t^i\|_2} \right\|_2, \quad (5)$$

where I indicates the indices of all teacher-student layers, which are then used to transfer the attention knowledge. Q_s^i and Q_t^i indicates the i -th student and teacher attention maps in vectorized forms, respectively, which can be calculated by using the attention mapping functions:

$$Q_s^i = \text{vec} \left(\sum_C |A_s^i|^a \right), \quad (6)$$

$$Q_t^i = \text{vec} \left(\sum_C |A_t^i|^a \right), \quad (7)$$

where $\sum_C$ indicates the accumulation along the channel dimension. a indicates norm type, and here it is empirically set to 2 for good activation-based attention knowledge transfer. As shown in Equations (6) and (7), the attention map generation method is very simple, that is, multi-channel feature maps are reduced along the channel dimension to generate a single-channel feature map. Therefore, the channel-wise knowledge will be ignored inevitably. Since we aim to learn the student network by making full use of the attention knowledge in the middle layers of the teacher network, we explore the attention knowledge of the teacher network from more aspects and transfer it to the student network.

3.4 Channel-based Attention Knowledge Transfer

Considering that the activation-based attention knowledge does not retain the knowledge of the channel dimension of feature representations, we thus transfer the channel-based attention knowledge during knowledge distillation in a similar way to [6, 34]. By doing this, student network can further learn the knowledge in different channels/layers of the teacher network. High-level attention from deep neural networks plays an important role in improving the model performance: when we transfer the activation-based attention knowledge at middle/deep layers, we also introduce the channel-based attention knowledge at the high level for knowledge distillation. As a result, we are able to not only improve the model performance but also reduce the training time. We describe the loss function for channel-based attention knowledge transfer as follows:

$$L_{CKT} = \sum_{i \in I} \left\| 1 - \frac{E_t^i}{\|E_t^i\|} * \frac{E_s^i}{\|E_s^i\|} \right\|_2^2, \quad (8)$$

where $E_s^i = G(F(A_s^i))$ and $E_t^i = G(A_t^i)$ indicate the student and teacher features after the global average pooling layer, respectively. G indicates the global average pooling operation, and F is a 1×1 convolutional transformation function, which is a linear mapping to bridge the difference between the channel dimensions of the teacher and student network. In fact, the loss of channel-based attention knowledge transfer is the cosine similarity loss $(1 - \cos\theta)^2$ with range values $[-1, 1]$, which can make the training loss to be well converged [34]. Therefore, within the proposed HMAT, channel-based attention knowledge can be efficiently transferred between different deep neural networks (especially deep neural networks with different numbers of channels in the middle layers).

3.5 HMAT Loss

Each attention mechanism has its own shortcomings in the process of KD. To improve the effectiveness of knowledge transfer, HMAT integrates multiple attention knowledge in a unified framework so that the network can focus on different aspects in different intermediate layers, mimicking the progressive process of humans to perceive the environment. The multi-attention loss function for knowledge distillation is formulated as follows:

$$L_{MAT} = \alpha L_{AKT} + \beta L_{CKT} + \gamma L_{PKT}. \quad (9)$$

In Equation (9), α , β and γ are the hyper-parameters of regularizing three types of the attention knowledge transfer, and they are empirically fine-tuned according to the corresponding related works. Therefore, the overall loss function for training very efficient student models can be defined as follows:

$$L_{total} = L_{MAT} + L_{task} + L_{KL}, \quad (10)$$

where L_{task} indicates the task-specific loss function, e.g., for image classification, it usually refers to the cross-entropy loss function, and for metric learning-based image retrieval, it may refer to the contrastive loss or the triplet loss. L_{KL} indicates the KL divergence between the outputs of the teacher network and the outputs of the student network. The loss function of KL divergence is generally formulated as

$$L_{KL} = \sum_{i=1}^n \sum_{j=1}^m p_t^{ij} \log \frac{p_t^{ij}}{p_s^{ij}}, \quad (11)$$

where n is the number of training samples from m classes, and p_s^{ij} and p_t^{ij} are the soft logit of the student and teacher networks for the i -th training sample associated with j -th class.

ALGORITHM 1: The proposed HMAT

Input: Data samples X with labels Y , student network S , teacher network T , hyperparameters α , β , γ , learning rate η .

- 1: **Initialize:** Initialize student network S , $epoch = 0$.
- 2: **Repeat:**
- 3: $epoch \leftarrow epoch + 1$, update η ;
- 4: A batch of data is randomly extracted from the training samples, and input into the student network and teacher network.
- 5: Compute position-based attention knowledge of student network and teacher network at the appropriate layers by Equation (4);
- 6: Compute activation-based attention knowledge of student network and teacher network at the appropriate layers by Equation (5);
- 7: Compute channel-based attention knowledge of student network and teacher network at the appropriate layers by Equation (8);
- 8: Compute final loss function by Equation (9);
- 9: Calculate gradients and update network's parameters.
- 10: $S \leftarrow S + \eta * \frac{\partial L_{total}}{\partial S}$.
- 11: **Until:** Converge.
- 12: **return** S .

Within the proposed framework for teacher-student learning, multiple attention knowledge transfer can provide more informative guidance to overcome the limitation of previous single attention knowledge transfer methods [21, 23, 52], and the experimental results can be found in Section 4. It is noteworthy that the proposed framework does not transfer all kinds of attention knowledge from all middle layers, but to selectively transfer different kinds of suitable attention knowledge from appropriate middle layers. Therefore, the proposed HMAT can effectively make full use of the advantages of different attention knowledge in a hierarchical manner, leading to an improved model performance. We summarize the overall training algorithm of our proposed HMAT framework in Algorithm 1.

4 EXPERIMENTS

In this section, we perform experiments to evaluate the proposed HMAT for knowledge distillation, including image classification, object detection, and image retrieval. Specifically, we conduct extensive experiments using the following datasets: (1) **image classification:** CIFAR-10 [25], CIFAR-100 [25], ImageNet [8]; (2) **object detection:** Pascal VOC [9, 10]; and (3) **image retrieval:** CUB-200-2011 [42], Cars 196 [24], Stanford Online Products [32]. For fair comparison, we choose peer networks from ResNet [15], MobileNet [37], ShuffleNet [29], and WideResNet (WRN) [48]. For ResNet, it is automatically divided into four sections according to ResBlocks in its network structure. For other networks, according to the structure of the network, we select four layers to output features. The proposed method, HMAT, is then compared to recent state-of-the-art methods for knowledge distillation, including AT [23], H-AT [34], CCKD [33], VID [2], CRD [41], and KDAFM [21]. Note that when referring to the competing methods, the performances are based on our reproduced implementation.

4.1 Datasets and Settings

CIFAR-10 and CIFAR-100: CIFAR-10 contains 60,000 images, including 50,000 images for training and 10,000 images for testing. All images are in the resolution of 32×32 pixels. All images are

from 10 different classes, where each class has the same number of samples for training and testing. During training, we use the following data augmentation: (1) all images are randomly cropped from zero-padded 40×40 pixels to 32×32 pixels; and (2) horizontal flip for mini-batch images with the probability 0.5. All models are trained for 200 epochs with the batch size 128 and the initial learning rate starts from 0.1 which is then reduced to its 1/5 at 60, 120, and 160 epochs.

ImageNet: ImageNet contains 1.2 million training and 50,000 testing images. All images are 64×64 pixels from 200 classes, and all classes has the same number of samples. During training, all images are randomly cropped and resized to 224×224 pixels. All models are trained for 90 epochs with the batch size 128 and the initial learning rate starts from 0.1 which is then reduced to its 1/10 at 30, 60, and 80 epochs.

CUB-200-2011 and Cars 196: CUB-200-2011 contains 5,994 training and 5,794 testing images from 200 different classes, while Cars 196 contains 5,144 training and 8,041 testing images from 196 classes. During training, we randomly crop zero-padded images from 256×256 pixels to 224×224 pixels and perform horizontal flip with the probability 0.5. All models are trained for 80 epochs with the batch size 128 and the initial learning rate starts from $1e-4$ which is then multiplied by 0.1 at 40 and 60 epochs.

Stanford Online Products: Stanford Online Products contains 59,551 training and 60,502 testing images from 22,634 classes. During training, we randomly crop zero-padded images from 256×256 pixels to 224×224 pixels. All models are trained for 120 epochs with the batch size 128 and the initial learning rate starts from $1e-4$ which is then multiplied by 0.1 at 40 and 80 epochs.

Pascal VOC: Pascal VOC2007 contains a *trainval* set with 5,011 images, and a *test* set with 4,952 images. Pascal VOC2012 contains a *trainval* set with 11,540 images, and a *test* set with 11,540 images. All images are from 20 different classes. During training, the student network is trained on the mixture of *trainval* sets from VOC2007 and VOC2012, and is then evaluated on the *test* set of VOC2007, which has been widely used as the object detection benchmark [19, 36]. All models are trained for 240,000 iterations with the batch size 24 and the initial learning rate starts from $1e-3$ which is then multiplied by 0.1 at 160,000 and 200,000 iterations.

For CIFAR-100 and Pascal VOC, we use SGD optimizer with the momentum 0.9 and the weight decay $5e-4$. For ImageNet, we use SGD optimizer with the momentum 0.9 and the weight decay $1e-4$. For CUB-200-2011, Cars 196 and Stanford Online Products, we use Adam optimizer with the weight decay $1e-5$. For other hyper-parameters, we use $\alpha = 300$, $\beta = 20$, and $\gamma = 0.05$, which is determined by a small grid-search. In our experiments, we consider both two settings, i.e., teacher-student networks have the same (or different) architecture.

4.2 Image Classification

Results on CIFAR-10/100: As shown in Table 1, our proposed HMAT outperforms the recent state-of-the-art methods, including AT and H-AT, in terms of the Top-1 accuracy. Specifically, when using ResNet18 with the ResNet34 as the student and teacher models, respectively, the proposed HMAT increases the Top-1 accuracy on CIFAR-10 by 0.5% compared to KD, 0.43% compared to AT, 0.44% compared to H-AT, 0.45% compared to CCKD, 0.34% compared to VID, 0.46% compared to KDAFM and 0.47% compared to CRD. As shown in Table 1, when ResNet18 is used as the student network, using ResNet18 as the teacher network sometimes performs better than using ResNet34 as the teacher network (e.g., the CCKD method). This indicates that the teacher network with better performance will not necessarily improve the performance of the student model better. Due to the inherent performance gap between the teacher network and the student network, the student

Table 1. The Top-1 Accuracy (%) of HMAT (Ours) and Other Comparison Methods on CIFAR-10 Dataset

Teacher	ResNet34	ResNet101	ResNet18	WRN-28-10	ResNet18	ShuffleNetV2
Student	ResNet18	ResNet50	ShuffleNetV2	WRN-16-2	ResNet18	ShuffleNetV2
Baseline_T	95.39	95.55	95.01	95.98	94.93	91.03
Baseline_S	94.93	95.10	91.03	93.54	94.93	91.03
KD	95.03 (↑ 0.10)	95.19 (↑ 0.09)	91.80 (↑ 0.77)	93.63 (↑ 0.09)	94.96 (↑ 0.03)	92.19 (↑ 1.16)
AT	95.10 (↑ 0.17)	95.22 (↑ 0.12)	93.12 (↑ 2.09)	93.91 (↑ 0.37)	95.00 (↑ 0.07)	91.19 (↑ 0.16)
H-AT	95.09 (↑ 0.16)	95.12 (↑ 0.02)	91.77 (↑ 0.74)	93.57 (↑ 0.03)	94.97 (↑ 0.04)	91.04 (↑ 0.01)
CCKD	95.08 (↑ 0.15)	95.25 (↑ 0.15)	92.45 (↑ 1.42)	93.59 (↑ 0.05)	95.16 (↑ 0.23)	92.48 (↑ 1.45)
VID	95.19 (↑ 0.26)	95.28 (↑ 0.18)	92.03 (↑ 1.00)	93.70 (↑ 0.16)	95.17 (↑ 0.24)	92.41 (↑ 1.38)
KDAFM	95.07 (↑ 0.14)	95.26 (↑ 0.16)	–	93.84 (↑ 0.30)	95.01 (↑ 0.08)	–
CRD	95.06 (↑ 0.13)	95.13 (↑ 0.03)	91.95 (↑ 0.92)	–	94.99 (↑ 0.06)	91.51 (↑ 0.48)
HMAT	95.53 (↑ 0.60)	95.42 (↑ 0.32)	93.58 (↑ 2.55)	94.36 (↑ 0.82)	95.24 (↑ 0.31)	92.61 (↑ 1.58)

The symbol ↑ means the improvement over the baseline of the student network.

Table 2. The Top-1 Accuracy (%) of HMAT (Ours) and Other Comparison Methods on CIFAR-100 Dataset

Teacher	ResNet34	ResNet101	ResNet18	WRN-28-10	ResNet18	ShuffleNetV2
Student	ResNet18	ResNet50	ShuffleNetV2	WRN-16-2	ResNet18	ShuffleNetV2
Baseline_T	77.66	78.88	75.77	79.78	75.77	70.10
Baseline_S	75.77	78.55	70.10	67.12	75.77	70.10
KD	76.47 (↑ 0.70)	79.08 (↑ 0.53)	72.23 (↑ 2.13)	70.75 (↑ 3.63)	76.99 (↑ 1.22)	71.75 (↑ 1.65)
AT	77.05 (↑ 1.28)	78.64 (↑ 0.09)	71.57 (↑ 1.47)	72.20 (↑ 5.08)	75.83 (↑ 0.06)	70.26 (↑ 0.16)
H-AT	77.82 (↑ 2.05)	79.07 (↑ 0.52)	71.51 (↑ 1.41)	71.93 (↑ 4.81)	76.68 (↑ 0.91)	70.14 (↑ 0.04)
CCKD	77.22 (↑ 1.45)	79.10 (↑ 0.55)	72.16 (↑ 2.06)	70.68 (↑ 3.56)	77.33 (↑ 1.56)	71.75 (↑ 1.65)
VID	77.24 (↑ 1.47)	78.82 (↑ 0.27)	72.30 (↑ 2.20)	71.69 (↑ 4.57)	77.17 (↑ 1.40)	72.18 (↑ 2.08)
KDAFM	77.94 (↑ 2.17)	79.15 (↑ 0.60)	–	71.98 (↑ 4.86)	77.68 (↑ 1.91)	–
CRD	76.42 (↑ 0.65)	79.10 (↑ 0.55)	70.63 (↑ 0.53)	–	76.33 (↑ 0.56)	71.18 (↑ 1.08)
HMAT	78.86 (↑ 3.09)	79.74 (↑ 1.19)	72.98 (↑ 2.88)	73.28 (↑ 6.16)	78.24 (↑ 2.47)	72.41 (↑ 2.31)

The symbol ↑ means the improvement over the baseline of the student network.

network cannot understand the knowledge from the teacher network well. In this case, the larger teacher network has a negative effect on the performance improvement of the student model.

As shown in Table 2, we report the Top-1 accuracy of recent state-of-the-art methods using different backbone networks on CIFAR-100 dataset. Among those knowledge distillation methods, the proposed HMAT outperforms recent state-of-the-art methods with a large margin, which demonstrates the effectiveness of multi-attention knowledge transfer in a hierarchical manner. Specifically, when using ResNet18 with the ResNet34 as the student and teacher models, respectively, the proposed HMAT increases the Top-1 accuracy on CIFAR-100 by 2.39% compared to KD, 1.81% compared to AT, 1.04% compared to H-AT, 1.64% compared to CCKD, 1.62% compared to VID, 0.92% compared to KDAFM, and 2.44% compared to CRD. Moreover, through the experimental results in Tables 1 and 2, we can see that H-AT with two types of attention knowledge transfer performs worse than AT with only single attention knowledge transfer in some cases. For example, on the teacher-student architectures using WideResNet AT performs better than H-AT on both CIFAR-10 and CIFAR-100 datasets. The possible reason is that H-AT performs the unvaried attention-based knowledge transfer in all middle layers of the teacher-student architectures, failing to transfer the valuable attention knowledge from different middle layers. Therefore, the proposed HMAT uses a variety of attention knowledge in different middle layers when performing knowledge transfer from teacher to student models, and has achieved significant improvements, further confirming that the hierarchical framework is very effective for knowledge distillation.

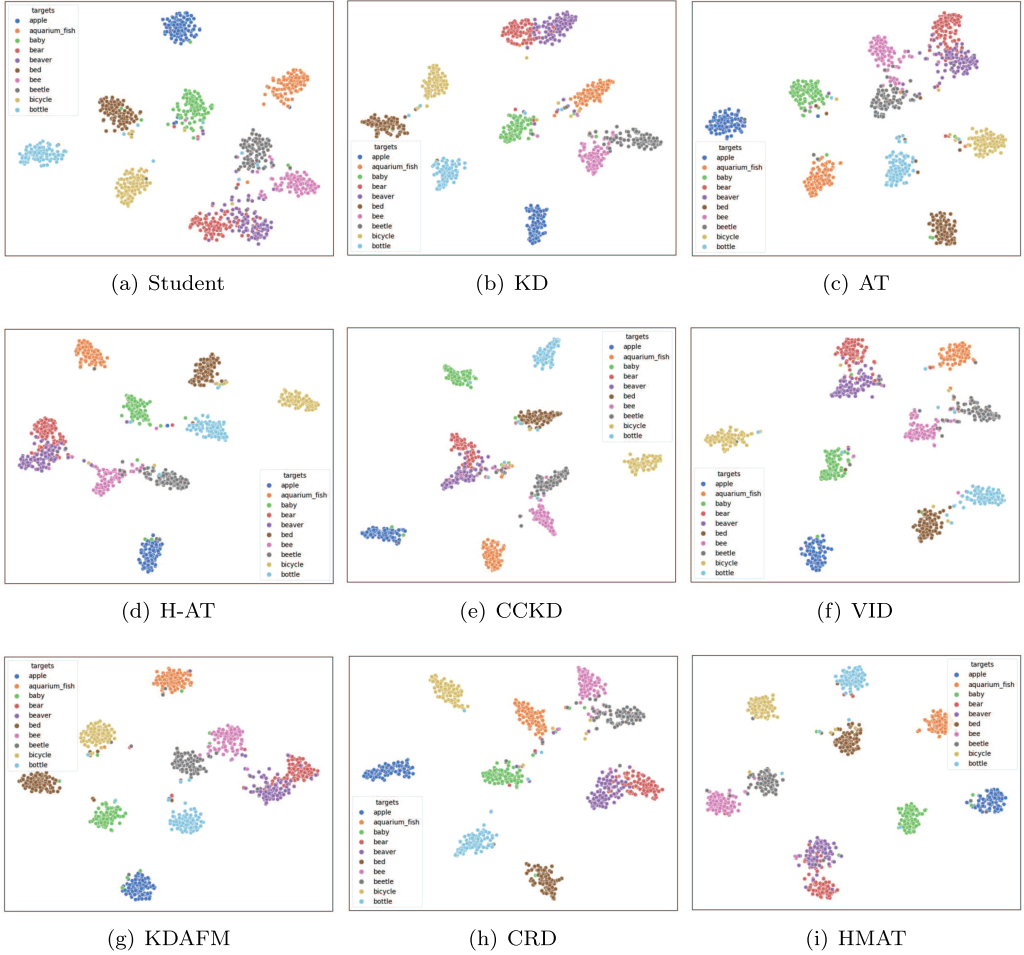


Fig. 5. t-SNE visualisation of ResNet18 trained with KD, AT, H-AT, CCKD, VID, KDAFM, CRD and HMAT on CIFAR-100 dataset. The teacher network we used here is ResNet34. Different colors indicate different classes. “Student” in (a) as the baseline refers to the student network (ResNet18) trained through the cross-entropy loss function.

To further demonstrate the advantages of our proposed method, we also use t-SNE to visualize the feature representations learned by different methods. Specifically, in Figure 5, we show the visualization results on CIFAR-100 dataset using KD, AT, H-AT, CCKD, VID, KDFAM, CRD, and the proposed HMAT. We find that our HMAT approach achieves more compact class-specific clusters than the others, which also demonstrates that the effectiveness of the proposed hierarchical multi-attention mechanisms or HMAT can distill more intermediate knowledge to facilitate the learning of the student model.

Results on ImageNet: We show the average Top-1 accuracy of recent state-of-the-art methods on ImageNet in Table 3. According to the experimental results, we can see that the proposed HMAT significantly outperforms all other methods. The significant improvements of HMAT comparing with H-AT and AT also indicates that the student model can learn more comprehensive attention knowledge from the teacher model when using a hierarchical framework. Therefore, with the

Table 3. The Top-1 Accuracy (%) of HMAT (Ours) and Other Comparison Methods on ImageNet Dataset

Teacher	Student	KD	AT	H-AT	CCKD	VID	KDAFM	CRD	HMAT
ResNet34	ResNet18	69.60	69.88	70.09	69.81	69.71	70.11	69.78	70.66
		73.06	69.36	($\uparrow$ 0.24)	($\uparrow$ 0.52)	($\uparrow$ 0.73)	($\uparrow$ 0.45)	($\uparrow$ 0.35)	($\uparrow$ 0.75)

The symbol $\uparrow$ means the improvement over the baseline of the student network.

proposed HMAT framework, we further explore the effectiveness of different kinds of attention knowledge in a unified framework, making it very different from the competing methods.

4.3 Object Detection

With the success of knowledge distillation in image classification tasks, researchers have studied whether the knowledge gained from image classification tasks can be transferred to other important vision tasks, such as object detection. We thus compare those attention-based knowledge transfer methods on Pascal VOC, which has been widely used for common object detection. For fair comparison, we use the standard evaluation metric evaluation, i.e., mean average precision or mAP, and the chosen teacher and student networks are MobileNetV2. We demonstrate the performance of KD, AT, H-AT, and HMAT in Table 4. We report the experimental results obtained on all classes (the entire dataset) and individual class. Comparing with KD, AT, and H-AT, the performance of our proposed HMAT increases by 9.65%, 0.59%, and 0.94%, respectively. In addition, our HMAT achieves the best performance in most classes (13 out of 20). As a result, our HMAT consistently outperforms other methods, demonstrating its effectiveness for downstream visual recognition tasks.

4.4 Image Retrieval

We evaluate the proposed HMAT method on metric learning-based image retrieval. The main idea of metric learning is to learn an embedding space which maximizes the inter-class variations and minimizes the intra-class variations. Specifically, we perform experiments on three popular image retrieval benchmarks including CUB-200-2011, Cars 196, and Stanford Online Products. In the experiments, ResNet34 and ResNet18 are respectively used as the teacher and the student during distillation process. For fair comparison, we use the standard evaluation metric, Recall@K. During testing, each image is first regarded as a query, and top K nearest neighbor images that do not contain the query are then retrieved from the *test* set: if any of retrieved images belongs to the same category with the query image, the Recall of this query is defined as 1.0. The final Recall@K is calculated as the average recall on the entire *test* set. As shown in Figure 6, the proposed HMAT outperforms recent state-of-the-art methods on three datasets with a clear margin when using Recall@1. Moreover, we visualize the attention maps of the middle layers of the neural network in Figure 7, where the columns indicate the original images and the visualization results of the block 0, block 1, block 2, and block 3 of deep neural networks, respectively. As shown in Figure 7, we find that low-level layers of deep neural networks usually pay more attention to the edges of the target objects, while the high-level layers usually pay more attention to the abstract information of the whole objects. The above observations inspire us to devise the multi-attention knowledge transfer for knowledge distillation in a hierarchical manner. Furthermore, on the Cars 196 dataset, our HMAT achieves an improvement of 3.63% over AT and 4.77% over H-AT. Though all three methods, AT, H-AT, and HMAT, transfer the attention knowledge from the teacher network to the student network, the performance of HMAT significantly outperforms the other two methods. We attribute the good performance of HMAT to the hierarchical structure for multiple attention knowledge transfer. Specifically, through the proposed HMAT, the student network learns the

Table 4. Experiments Results (%) on the Pascal VOC Dataset

MobileNetV2	KD	AT	H-AT	HMAT
VOC	62.25	71.31	70.94	71.90
Person	69.30	74.00	73.29	<u>74.67</u>
Aeroplane	66.83	72.18	71.47	<u>73.43</u>
Bicycle	71.47	78.81	79.23	<u>81.46</u>
Motorbike	72.73	79.61	80.35	<u>81.60</u>
Car	74.79	78.91	78.48	<u>79.10</u>
Bus	70.60	<u>81.25</u>	79.43	80.71
Boat	47.15	<u>61.73</u>	59.02	60.83
Train	77.44	<u>83.68</u>	82.19	83.00
Cat	80.42	83.97	83.81	<u>85.50</u>
Dog	76.73	80.51	81.51	<u>81.67</u>
Cow	56.46	72.10	<u>72.98</u>	72.31
Bird	61.14	<u>69.24</u>	68.36	68.28
Horse	80.30	84.21	83.85	<u>84.39</u>
Sheep	56.51	69.85	68.77	<u>70.96</u>
Chair	45.00	54.24	53.18	<u>54.74</u>
Dining Table	52.89	73.16	72.06	<u>73.51</u>
Sofa	62.21	78.43	<u>79.28</u>	78.12
Bottle	27.94	36.27	<u>36.49</u>	36.19
TV/monitor	62.15	67.94	69.24	<u>69.48</u>
Potted Plant	32.97	46.07	45.76	<u>48.08</u>

The first line is the mAP results obtained by MobileNetV2 on all classes (the entire VOC dataset) through the four compared methods, and the other lines are the mAP results obtained by MobileNetV2 on each individual class through the four compared methods. We underline the best performance of the current class.

position-based attention knowledge in early layers of the teacher network at the low level, and learns the activation-based and channel-based attention knowledge about abstract information in middle/deep layers of the teacher network at the high level.

4.5 Ablation Studies

To understand how each attention knowledge contributes to the good performance and explore the effectiveness of HMAT for knowledge transfer, we introduce six variants of HMAT as follows. It should be noted that here our proposed HMAT model is also regarded as the variant A for simple ablation comparison.

- Variant A: The full strategy (i.e., HMAT) using the three kinds of attention knowledge transfer altogether during distillation. The loss function used in this case is Equation (10);
- Variant B: In the distillation process, activation-based attention knowledge transfer (AKT) and channel-based attention knowledge transfer (CKT) are carried out. In this case, this variant is the same as H-AT. The formula of the loss function is $\alpha L_{AKT} + \beta L_{CKT} + L_{CE}$;
- Variant C: In the distillation process, activation-based attention knowledge transfer (AKT) and position-based attention knowledge transfer (PKT) are carried out. The formula of the loss function is $\alpha L_{AKT} + \gamma L_{PKT} + L_{CE}$;
- Variant D: In the distillation process, channel-based attention knowledge transfer (CKT) and position-based attention knowledge transfer (PKT) are carried out. The formula of the loss function is $\beta L_{CKT} + \gamma L_{PKT} + L_{CE}$;

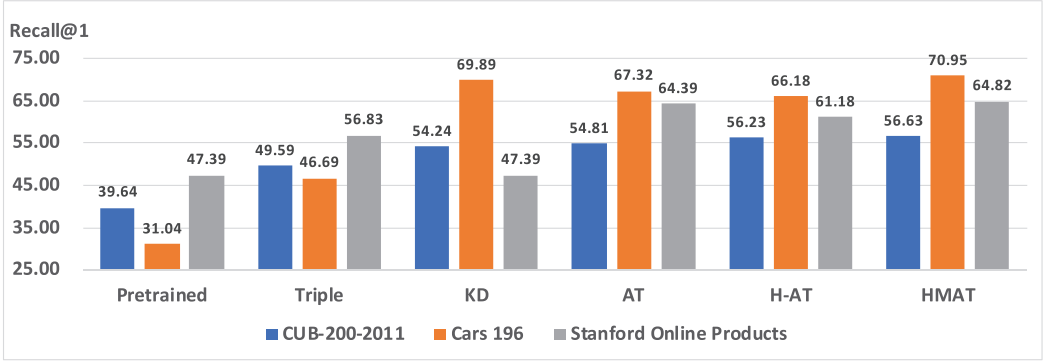


Fig. 6. Experimental results on three image retrieval datasets, CUB-200-2011, Cars 196, and Stanford Online Products. Specifically, we report the performance using Recall@1. Among them, Pretrained indicates the performance of the untrained student model, and Triple indicates the performance of the trained student network.

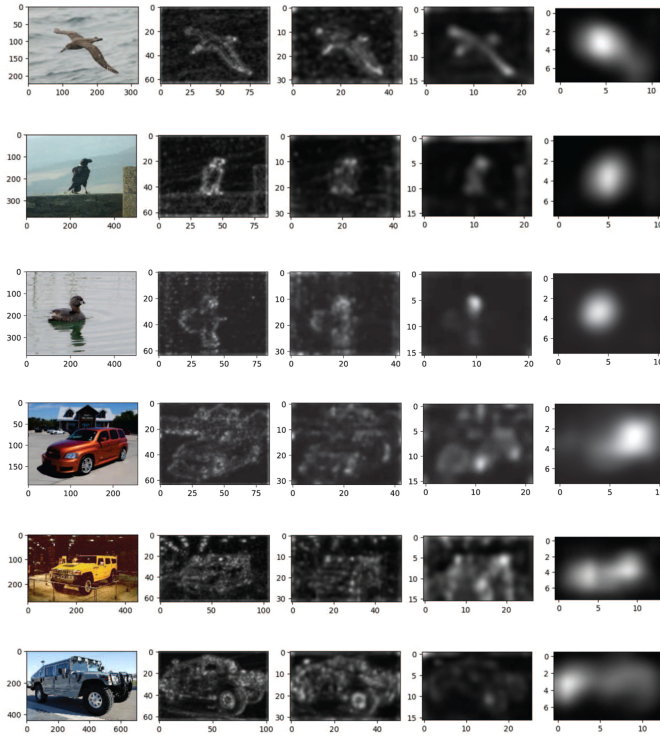
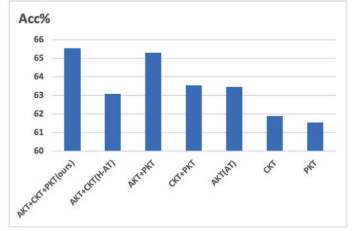
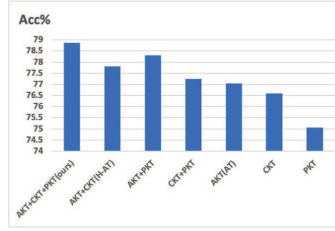
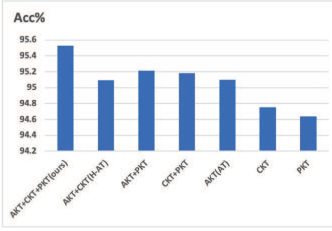


Fig. 7. A visualization of the attention maps from the middle layers of deep neural network. Specifically, the first three rows show images from the CUB-200-2011 dataset and the last three rows show images from the Cars 196 dataset. We use ResNet18 as the backbone network here. The columns from left to right indicate the original picture, the output of the first, second, third, and fourth blocks of deep neural networks, respectively.

Table 5. The HMAT Variants with/without Each Attention Knowledge Transfer

	AKT	CKT	PKT
Variant A	✓	✓	✓
Variant B	✓	✓	✗
Variant C	✓	✗	✓
Variant D	✗	✓	✓
Variant E	✓	✗	✗
Variant F	✗	✓	✗
Variant G	✗	✗	✓



(a) HMAT variants on CIFAR-10 (b) HMAT variants on CIFAR-100 (c) HMAT variants on Tiny-ImageNet

Fig. 8. Ablation studies on three different datasets, including CIFAR-10, CIFAR-100, and Tiny-ImageNet. Each subfigure contains our HMAT method and the variants of HMAT, each of which excludes one or two attention knowledge from the full strategy. Here we choose ResNet34 as the teacher network and ResNet18 as the student network.

- Variant E: Only activation-based attention knowledge is transferred during distillation process. This variant is the AT scheme, i.e., $\alpha L_{AKT} + L_{CE}$;
- Variant F: Only channel-based attention knowledge is transferred during distillation process. The loss function of this variant is $\beta L_{CKT} + L_{CE}$;
- Variant G: Only position-based attention knowledge is transferred during distillation process. The loss function of this variant is $\gamma L_{PKT} + L_{CE}$.

Among the variants, L_{AKT} , L_{CKT} , and L_{PKT} respectively correspond to Equations (5), (8), and (4), and L_{CE} is the traditional cross-entropy function for classification tasks. Note that the hyperparameters α , β , and γ used in these variants are similarly set as in the proposed method. Variant A is to transfer three kinds of attention knowledge in the four layers selected in the network, Variant B, C, and D refers to transfer two kinds of attention knowledge in four layers selected in the network, Variant E, F, and G refers to transfer one kind of attention knowledge in four layers selected in the network. These variants are intuitively shown in Table 5.

Figure 8 shows the Top-1 accuracy of the above seven variations on the CIFAR-10, CIFAR-100 and Tiny-ImageNet datasets. Note that here we use a subset of ImageNet, i.e., Tiny-ImageNet, to fastly conduct the ablation experiments. From these ablation experimental results, we then have:

- Variant A outperforms all other variants (where one or two attention knowledge has been removed during distillation training). It means that the use of multiple attention knowledge plays a crucial role in knowledge distillation. In the process of knowledge transfer, different attention knowledge is transferred in different middle layers of neural network, which can

make the student network acquire proper knowledge from each layer of the teacher network and simultaneously make the acquired knowledge more comprehensive.

- The performance will decrease when removing the attention knowledge that can be transferred during the distillation training process, except variant B (H-AT). This phenomenon further demonstrates that simply transferring multiple attention knowledge among all intermediate layers without considering different layers may cause negative effects.
- AT (Variant E) and H-AT (Variant B) can be seen as special cases of HMAT with only one or two kinds of attention knowledge during distillation. However, our method transfers different attention knowledge in different middle layers of the network, which is different from the idea of the above-mentioned methods.

Attention mechanisms have turned out to be very effective for knowledge transfer [21, 23, 34]. Nevertheless, there is a trade-off between knowledge and noise during knowledge transfer, where the ablation studies are provided in Figure 4 to demonstrate the effectiveness of the semantic knowledge of different layers in a hierarchical attention manner.

5 CONCLUSION

In this paper, we devise a novel knowledge distillation method called hierarchical multi-attention transfer or HMAT for knowledge distillation. The proposed HMAT naturally integrates different kinds of attention knowledge in a unified framework so that attention knowledge can be transferred effectively in-between different teacher-student architectures. Experimental results on seven datasets from three important visual recognition tasks have verified that the proposed HMAT outperforms the recent attention-based knowledge transfer methods with a large margin, demonstrating the effectiveness of transferring multiple types of attention knowledge simultaneously in a hierarchical manner. We hope that the proposed method can provide some insights on attention-based knowledge transfer to facilitate the future works on knowledge distillation. In the future work, we will plan to extend the proposed hierarchical multi-attention knowledge transfer by utilizing each attention knowledge with the adaptively learned weight to be distilled at different layers during the distillation training process.

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